



**AIN SHAMS UNIVERSITY**  
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**Major Task Report**

***Submitted to:***

Prof. Sameh A. Ibrahim

***Submitted by:***

Manar Saber Abdelrahim 22P0125

Mariam Islam Elsebaie 22P0177

Ahmed Sherif Mohamed 23P0414

# Table of Contents

|                                           |    |
|-------------------------------------------|----|
| 1- Introduction .....                     | 3  |
| 2- LUT Generation.....                    | 5  |
| 3- Telescopic Cascode Design.....         | 10 |
| 4- Frequency Response Hand Analysis ..... | 15 |
| 5- DC Cascode Simulation.....             | 19 |
| 5.1 Schematic: .....                      | 19 |
| 5.2 Setting Output Voltage value:.....    | 20 |
| 5.3 DC Operating point (OP): .....        | 21 |
| 6- AC Cascode Simulation.....             | 24 |
| 7- Phase Margin: .....                    | 25 |
| 8- Corner and Monte Carlo Simulation..... | 26 |
| 8.1 Corners Simulation: .....             | 26 |
| 8.2 Monte Carlo Simulation:.....          | 26 |

# 1- Introduction

Telescopic cascode differential amplifiers are widely used in analog integrated circuits due to their high voltage gain, low power consumption, and simple topology. These amplifiers are particularly suitable for low-voltage applications where achieving high gain with limited power budget is required. In modern CMOS technologies, careful biasing and device sizing are essential to ensure proper operation and compliance with design constraints.

In this project, a telescopic cascode differential amplifier is designed and simulated using TSMC 65-nm CMOS technology with a 1.2 V supply voltage. The design targets a minimum voltage gain of 500 while maintaining power consumption not exceeding 3 mW. Hand analysis is first performed to determine transistor dimensions, bias currents, and operating regions, followed by circuit-level simulations to verify performance.

Simulation results are used to confirm that all transistors operate in saturation, the required voltage gain is achieved, and the frequency response meets expectations. Finally, a comparison between analytical calculations and simulation results is presented to evaluate the accuracy of the design methodology and to account for the effects of short-channel behavior in deep-submicron CMOS technology.

## ***$g_m/I_D$ Design Methodology:***

It is a simulation-driven design approach that focuses on the transconductance efficiency of MOSFETs as a key design variable. Unlike the square-law model, which fails in deep-submicron technologies, this method provides a more accurate and intuitive understanding of the trade-offs between transistor sizing, operating region, power consumption, and gain

## ***Key Features:***

- **Device Characterization:** The  $g_m/I_D$  ratio is extracted from simulation-based LUTs, representing how effectively a transistor converts bias current into transconductance.
- **Region Selection:** The method allows designers to target specific inversion regions (weak, moderate, or strong) depending on design goals like noise, speed, or power.
- **Analytical Design:** By relating  $g_m/I_D$  to overdrive voltage  $V_{ov}$  output resistance  $r_o$  and intrinsic gain  $g_m r_o$ , it bridges simulation data with classical small-signal design equations.

## ***How It Was Used in This Project:***

1. **LUT Generation:** DC sweeps were performed to generate LUTs for NMOS and PMOS devices showing curves of  $g_m/I_D$ ,  $g_m r_o$  and  $I_D/w$  across different L values.
2. **Sizing and Biasing:** Transistor lengths and bias points were chosen based on the desired  $g_m/I_D$  values, ensuring operation in the appropriate inversion level for optimal performance.
3. **Gain Maximization:** The selected  $g_m/I_D$  and  $r_o$  values were used into the

small-signal gain equation to achieve the required differential gain  $> 500$ .

4. **Power Optimization:** The method allowed control over bias current for each branch, ensuring the total power stayed below 3 mW.

This approach offered high accuracy, better design insight, and robust simulation alignment, making it particularly suitable for the TSMC 65nm process used in this project.

## 2-LUT Generation

This step characterizes the 65 nm TSMC NMOS and PMOS devices by generating simulation-based Lookup Tables (LUTs). The LUTs relate biasing conditions to key performance parameters such as transconductance efficiency, overdrive voltage, and intrinsic gain, enabling accurate device sizing and operating-point selection.

This approach replaces inaccurate analytical equations with simulation-derived data while preserving classical analytical circuit design methods. As a result, it combines high accuracy with strong design intuition.

### Square Law:

In classical analog circuits, the MOSFET is modeled using the **square-law equation**.

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

However, in modern deep submicron MOSFET technologies, this model becomes **increasingly inaccurate** due to **short-channel effects**, causing the following effects.

- The quadratic dependence on  $V_{GS}$  no longer holds. Hence, the error in hand calculations increases significantly.
- Device performance parameters like intrinsic gain and  $V_{TH}$  are no longer easily predictable.

As a result of the invalidity of square law,  $V_{ov}$  loses its accuracy and usefulness in design. Hence, another design variable is needed:  $V^*$ . This parameter, also known as the effective overdrive, directly relates to the transistor's operating region and performance. It can be extracted from LUTs by equation below. In other words, LUTs will serve as accurate references throughout the circuit design process.

$$V^* = 2 \frac{I_D}{g_m}$$

Based on those concepts, the design mainly adopted  $\frac{g_m}{I_D}$  **design methodology** (offers insight into the device's inversion level and performance).

### LUT Extraction:

LUTs for both NMOS and PMOS devices were generated under the same simulation conditions (which are summarized below). There were Two LUTs that are generated: ( $g_m r_o$  (intrinsic gain),  $I_D/w$  (Normalized current)).

#### 1. $V_{DS} = 0.2 V$ & $0.3 V$

As it has a **secondary impact** on **device parameters**, it was chosen roughly as 0.2 V for some transistors and 0.3 V for the others (Having 1.2 V disturbed upon 5 transistors and rounding down for margin).

**2.  $V_{SB} = 0\text{ V}$**

The body-source voltage was set to zero to eliminate body effect contributions and **simplify design**.

**3. L (Swept over range  $0.1\text{ }\mu\text{m}$  till  $1\text{ }\mu\text{m}$ )**

Discrete values of channel length L were swept during simulation to **observe its effect** on **device performance**.

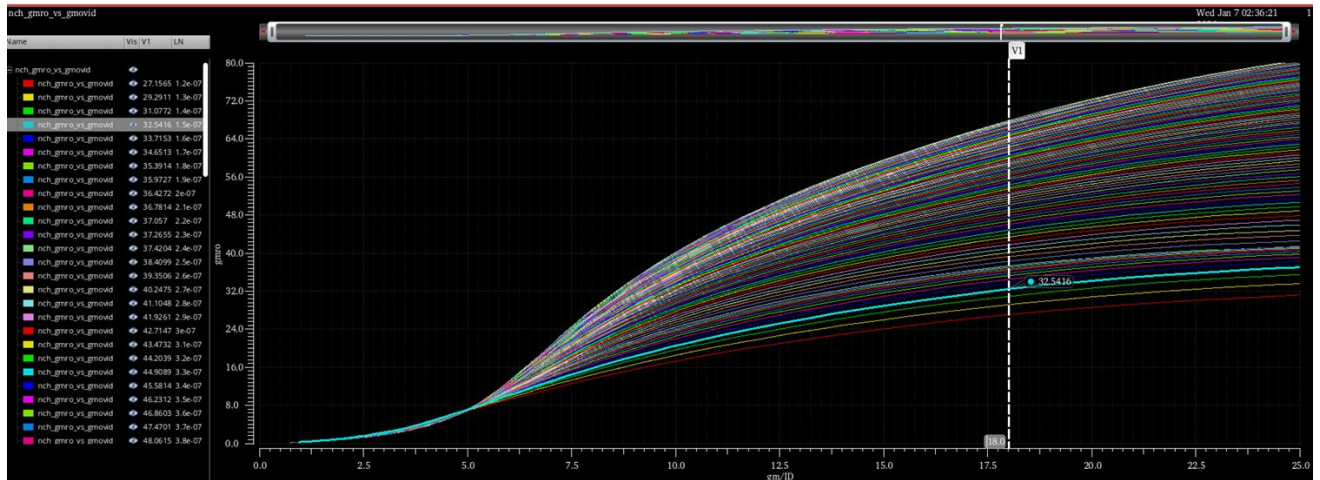
**4. W (Kept constant to  $1\text{ }\mu\text{m}$ )**

Transistor width was fixed to  $W_{ref} = 1\text{ }\mu\text{m}$  for all simulations.

## Charts:

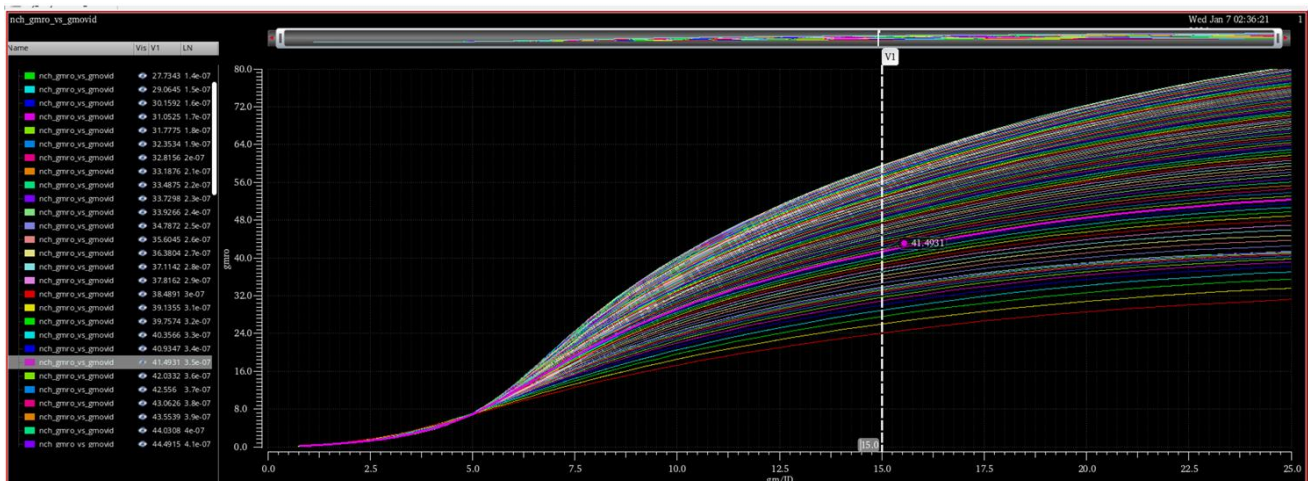
### NMOS: M1 & M2

$$g_m \times r_o:$$



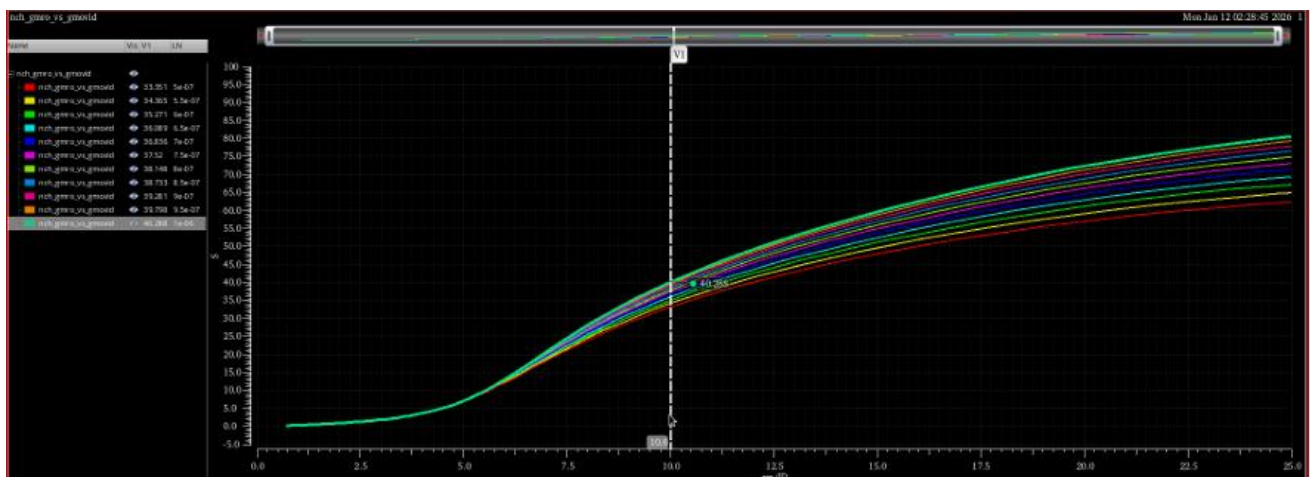
- $L1 = 0.15 \mu m$

### NMOS: M3 & M



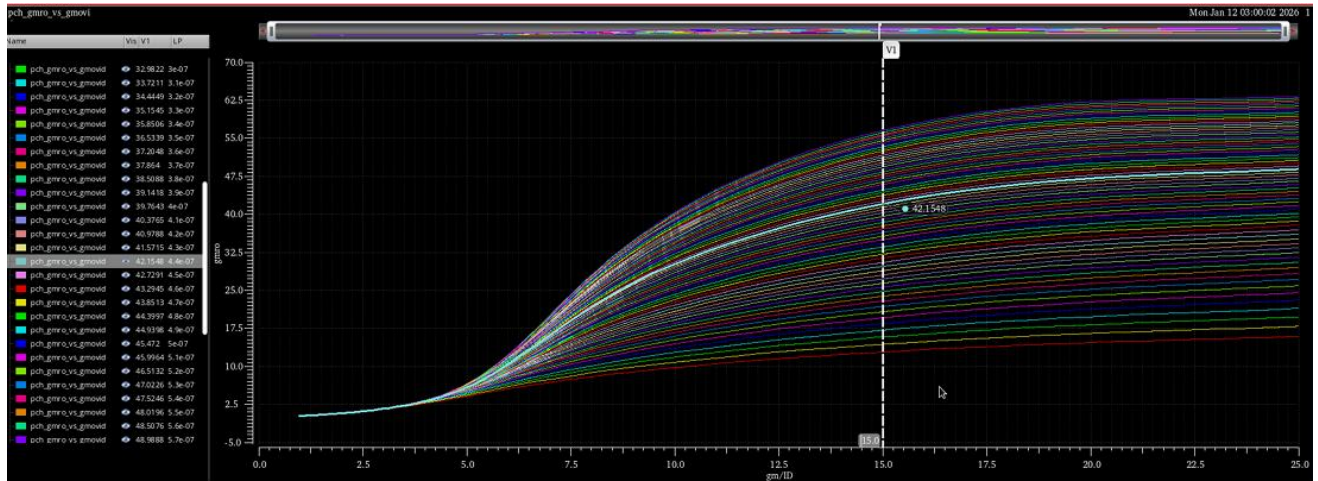
- $L3 = 0.35 \mu m$

### NMOS: M9(Current Source)



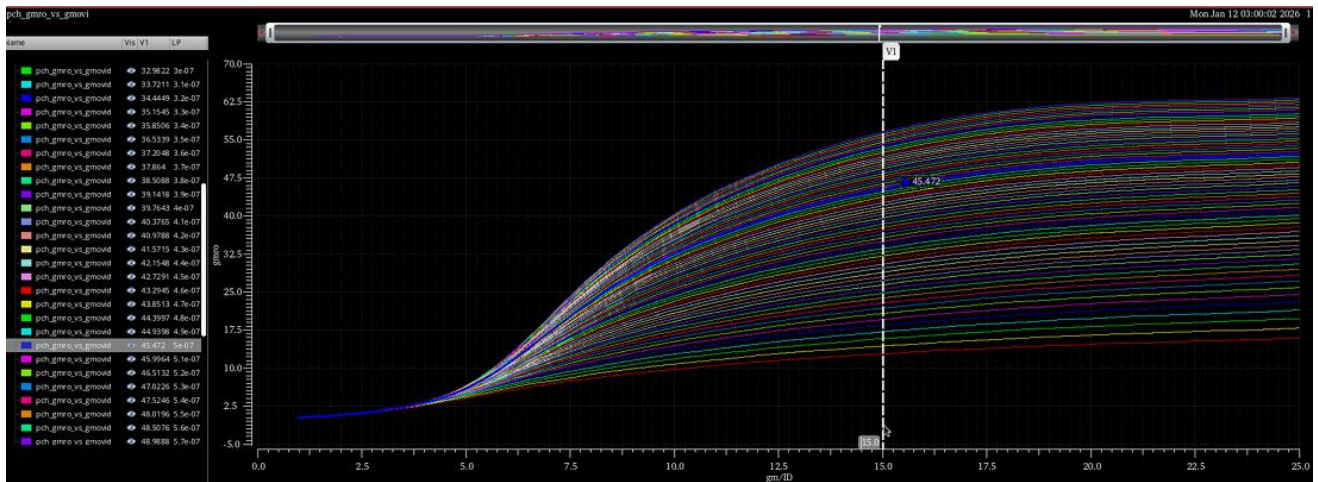
- $L9 = 1\ \mu m$

PMOS: M5 & M6



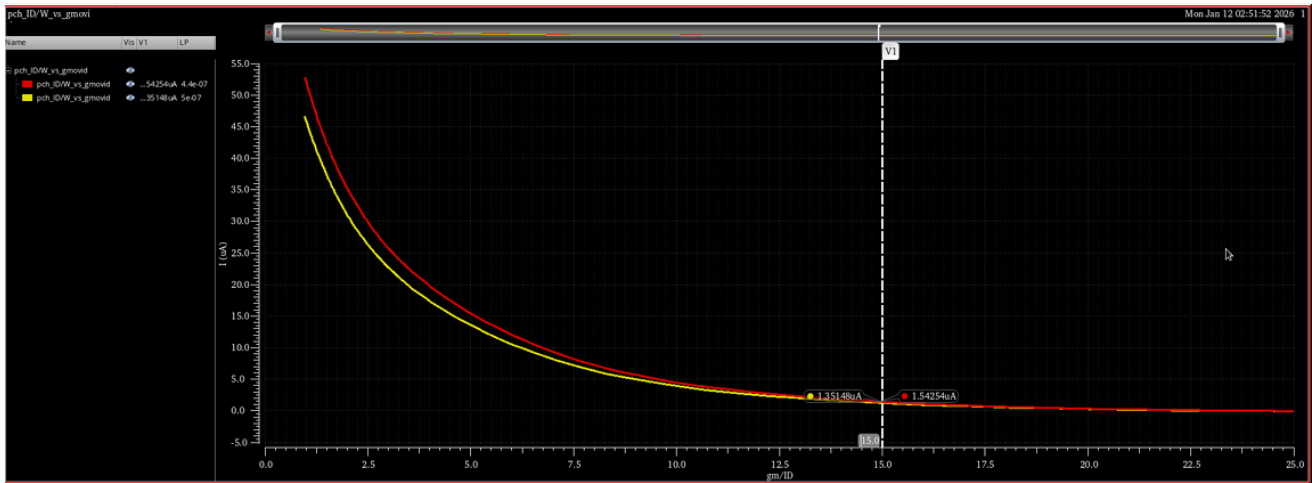
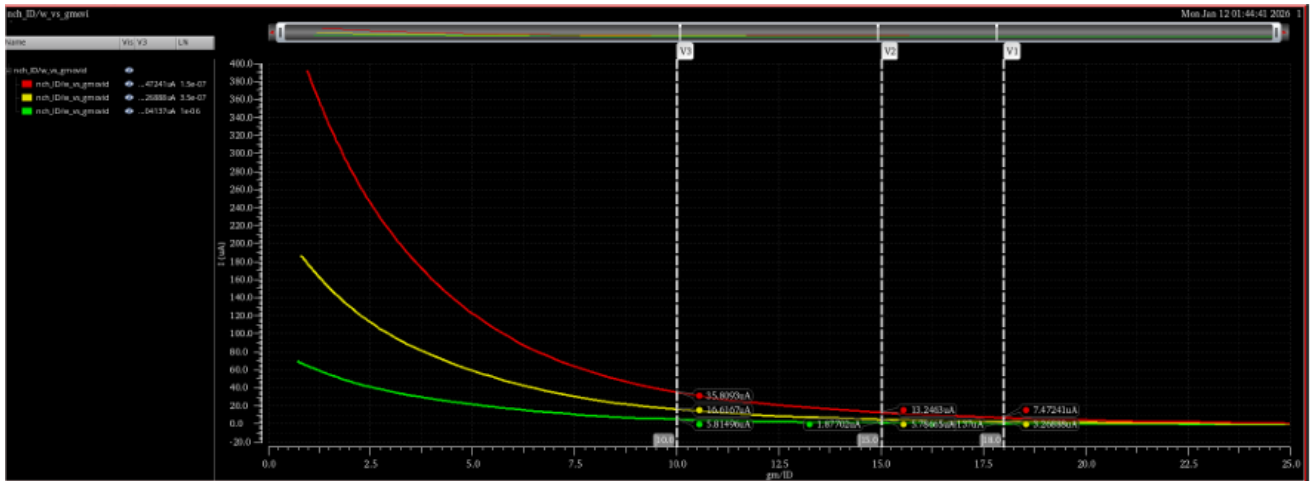
- $L5 = 0.44\ \mu m$

PMOS: M7 & M8



- $L7 = 0.5\ \mu m$

***ID/W:***



### 3-Telescopic Cascode Design

#### Specifications' limitations:

In this section, the absolute limitations of parameter using  $A_V$  and power are explored before jumping into design step. Moreover, some assumptions and the exact requirements are specified.

From **AV & Power**,

$$A_V = G_M \times R_{OUT} \geq 500 \quad \& \quad I_{ss} \leq 2.5mA$$

For Design approach, a margin was taken for  $A_V$ . Hence, the value of  $A_V$  was set to be 650.

**Design for  $A_V = 620$**

#### Setting Current:

Assuming the reference current  $I_{REF} = 80 \mu m$  is provided by a bandgap reference (BGR) circuit, the drain current  $I_D$  was chosen to match this value. Selecting,  $I_{ss} = I_{REF} = 80 \mu m$  minimizes total power consumption while ensuring consistency in current mirroring (a single instance which can be matched). Specifically,  $I_D = 40 \mu m$  was chosen to make half circuit current having integer valued current.

#### Gain Equation:

$$A_V = g_{m_{input\ pair}} (g_{m_{NMOS\ Active\ Load}} \times r_{o_{NMOS\ Active\ Load}} \times r_{o_{Input\ pair}} \parallel g_{m_{PMOS\ Active\ Load}} \times r_{o_{PMOS\ Active\ Load}} \times r_{o_{PMOS\ Current\ source\ Load}})$$

$$\therefore A_V = g_{m_{M1,2}} (g_{m_{M3,4}} \times r_{o_{M3,4}} \times r_{o_{M1,2}} \parallel g_{m_{M5,6}} \times r_{o_{M5,6}} \times r_{o_{M7,8}})$$

The transistors in the telescopic cascode amplifier can be categorized into three functional groups:

- **Transistors contributing to gain through both  $g_m$  and  $r_o$ :** M1, M2, M3, M4, M5, and M6
- **Transistors contributing to gain through  $r_o$  only:** M7 and M8
- **Transistor responsible for setting the biasing current:** M9

**Note:** Within the earlier categorization based on contributions to gain via  $g_m$  and  $r_o$ , the input differential pair (M1 and M2) holds a **distinct and more critical role**. Unlike other transistors, the input pair contributes to the gain not only through its output resistance  $R_{OUT}$  (using  $r_o$ ), but also **directly to the overall transconductance  $G_M$**  via its  $g_m$ . Since  $A_V = G_M \times R_{OUT}$ , the input pair influences both terms in the gain equation, giving it a **higher weight in determining the amplifier's performance** compared to the other devices. This further justifies its careful sizing and biasing in the design process.

Each group is biased differently based on its function and its contribution to the overall gain. To optimize performance while minimizing area, the biasing conditions are selected based on trends observed in  $\frac{g_m}{I_d}$  versus transistor performance charts: **Mainly** the  $g_m \times r_o$  vs  $g_m/I_d$ .

### Transistor Sizing:

| Global Variables                    |         |             |
|-------------------------------------|---------|-------------|
| <input checked="" type="checkbox"/> | IDC     | 200u*K      |
| <input checked="" type="checkbox"/> | VDD     | 1.2         |
| <input checked="" type="checkbox"/> | VIND    | 1           |
| <input checked="" type="checkbox"/> | VM34    | 0.95        |
| <input checked="" type="checkbox"/> | VM56    | 0.6         |
| <input checked="" type="checkbox"/> | VM78    | 808.5m      |
| <input checked="" type="checkbox"/> | A_CMFB  | -5          |
| <input checked="" type="checkbox"/> | LP_CASC | 440n        |
| <input checked="" type="checkbox"/> | LP_LOAD | 500n        |
| <input checked="" type="checkbox"/> | WP_CASC | 64.8283u*K  |
| <input checked="" type="checkbox"/> | WP_LOAD | 72.1829u*K  |
| <input checked="" type="checkbox"/> | LN_CASC | 350n        |
| <input checked="" type="checkbox"/> | LN_CM   | 1u          |
| <input checked="" type="checkbox"/> | LN_IN   | 150n        |
| <input checked="" type="checkbox"/> | WN_CASC | 17.2812u*K  |
| <input checked="" type="checkbox"/> | WN_CM   | 17.24427u*K |
| <input checked="" type="checkbox"/> | WN_IN   | 13.3826u*K  |
| <input checked="" type="checkbox"/> | VINCM   | 0.7         |
| <input checked="" type="checkbox"/> | VOUTCM  | 0.72        |
| <input checked="" type="checkbox"/> | K       | 0.4         |

Click to add variable

### Global Scaling Factor (K):

A global scaling factor of  $K=0.4$  was applied to both the transistor widths (W) and the bias currents (ID) to reduce power consumption while preserving the operating point. Since the current density ( $ID/W$ ) remains constant, the overdrive voltage (VOV) and transconductance efficiency ( $g_m/ID$ ) are unchanged. However, reducing ID increases the output resistance ( $r_o \approx 1/(\lambda * ID)$ ), which enhances the overall output resistance of the telescopic cascode. This allows the required high voltage gain to be achieved while keeping the total power consumption below the 3 mW limit.

$$I_D = 40 \mu A$$

$$\text{let } \frac{g_{m1}}{I_D} = 18 \rightarrow g_{m1} = 720 \mu S, V_{OV1} \approx V_1^* = 0.1111 V$$

$$\text{let } \frac{g_{m3}}{I_D} = \frac{g_{m5}}{I_D} = \frac{g_{m7}}{I_D} = 15 \rightarrow V_{OV3} = V_{OV5} = V_{OV7} \approx 0.1333 V$$

$$g_{m3} = g_{m5} = g_{m7} = 600 \mu S$$

$$\text{let } \frac{3}{2} r_{o1} = r_{o7} = r_{o3} = r_{o5}$$

$$|A_v| \approx g_{m1} (r_{o1} g_{m3} r_{o3} // r_{o5} g_{m5} r_{o7}) = 620$$

$$720 \mu S \left( \frac{3}{2} r_{o1}^2 \times 600 \mu S // \left( \frac{3}{2} r_{o1}^2 \times 600 \mu S \right) \right) = 620$$

$$r_{o1} = 40 K\Omega$$

$$r_{o3} = r_{o5} = r_{o7} = 60 K\Omega$$

NMOS

- $M_1: g_{m1} r_{o1} = 28.8$   
We will choose 150 nm which has  $g_m r_o = 32.5416$   
and  $I_D |_{W=1\mu} = 7.47 \mu A$   $r_{o1} = 45.2 K\Omega$   
 $W_1 = \frac{40 \mu A}{7.47 \mu A} = 5.355 \mu m$
- $M_3: g_{m3} r_{o3} = 36$   
We will choose 350 nm which has  $g_m r_o = 41.497$   
and  $I_D |_{W=1\mu} = 5.7866 \mu A$   $r_{o3} = 69.155 K\Omega$   
 $W_3 = \frac{40 \mu A}{5.7866 \mu A} = 6.9125 \mu m$
- $M_9: \text{let } \frac{g_m}{I_D} = 10 \rightarrow g_m = 400 \mu S, V_{OV9} \approx 0.2 V$   
let  $L = 1 \mu m$   
 $g_m r_o = 40.2877 \rightarrow r_o = 100.72 K\Omega$

$$I_D |_{W=1\mu} = 5.815 \mu A$$

$$W_9 = \frac{40}{5.815} = 6.879 \mu m$$

PMOS  $\rightarrow$   $M_5: g_{m5} r_{o5} = 36$

We will choose  $440\text{ nm}$  which has  $g_{m0} = 42.164\mu$

$I_D/W = 1.5425\mu$

$r_{o5} = 70.27\text{ k}\Omega$

$W_5 = \frac{40}{1.5425} = 25.932\mu\text{m}$

$\rightarrow M_7: g_{m7} r_{o7} = 36$

We will choose  $500\text{ nm}$  which has  $g_{m0} = 45.472$

$I_D/W = 1.351\mu\text{A}$

$r_{o7} = 75.79\text{ k}\Omega$

$W_7 = \frac{40}{1.351} = 29.607\mu\text{m}$

After deciding the lengths of our MOSFET, we can easily get the rest of parameters for each MOSFET from the LUT.

| Transistor               | NMOS Input pair (M1 & M2) | NMOS Active Loads (M3 & M4) | Tail current source (M9) |
|--------------------------|---------------------------|-----------------------------|--------------------------|
| L                        | $0.15\mu\text{m}$         | $0.35\mu\text{m}$           | $1\mu\text{m}$           |
| $R_o$                    | $45.2\text{ k}\Omega$     | $69.155\text{ k}\Omega$     | $100.72\text{ k}\Omega$  |
| ID                       | $40\mu\text{m}$           | $40\mu\text{m}$             | $80\mu\text{m}$          |
| Vov                      | $0.111\text{ V}$          | $0.1333\text{ V}$           | $0.2\text{ V}$           |
| Gm/ID                    | 18                        | 15                          | 10                       |
| Gm                       | $720\mu\text{s}$          | $600\mu\text{s}$            | $400\mu\text{s}$         |
| $X = \frac{id}{W_{ref}}$ | 7.47                      | 5.7866                      | 5.815                    |
| $W = \frac{id}{X}$       | $5.355\mu\text{m}$        | $6.9125\mu\text{m}$         | $6.879\mu\text{m}$       |

| Transistor               | PMOS Active Loads (M5 & M6) | PMOS current source (M7&M8) |
|--------------------------|-----------------------------|-----------------------------|
| L                        | $0.44\mu\text{m}$           | $0.5\mu\text{m}$            |
| $R_o$                    | $70.27\text{ k}\Omega$      | $75.79\text{ k}\Omega$      |
| ID                       | $40\mu\text{m}$             | $40\mu\text{m}$             |
| Vov                      | $0.133\text{ V}$            | $0.133\text{ V}$            |
| Gm/ID                    | 15                          | 15                          |
| Gm                       | $600\mu\text{s}$            | $600\mu\text{s}$            |
| $X = \frac{id}{W_{ref}}$ | 1.5425                      | 1.351                       |
| $W = \frac{id}{X}$       | $25.932\mu\text{m}$         | $29.607\mu\text{m}$         |

### DC Biasing voltages calculations:

- **For VICM:**

$$VICM = V_{gs1} + V_{ds9} = 0.419 + 0.281 = \mathbf{0.7\ V}$$

- **For  $V_{b1}$ :**

$$V_{b1} = V_{gs3} + V_{ds1} + V_{ds9} = 0.4516 + 0.2175 + 0.281 = \mathbf{0.95\ V}$$

- **For  $V_{b2}$ :**

$$V_{b2} = V_{dd} - V_{ds7} - V_{sg5} = 1.2 - 0.2101 - 0.3899 = \mathbf{0.6\ V}$$

- **For  $V_{b3}$ :**

$$V_{b3} = V_{dd} - V_{sg7} = 1.2 - 0.3915 = \mathbf{0.8085\ V}$$

- **For VOCM:**

$$VOCM = V_{ds1} + V_{ds3} + V_{ds9} = 0.2175 + 0.2216 + 0.281 = \mathbf{0.72\ V}$$

$$\begin{aligned} P_{total} = \sum P_i &= (40 \times 0.2 + 40 \times 0.2 + 40 \times 0.3 + 40 \times 0.3) \times 2 + 80 \times 0.2 \\ &= 96\ \mu\text{W} \end{aligned}$$

## 4-Frequency Response Hand Analysis

After adding the capacitance of each transistor to it, we found that:

- $C_{gs}$ ,  $C_{SB}$  of PMOS transistor M7 is grounded
- $C_{SB}$  of the NMOS transistor M1 is grounded too

And we will use miller approximation for  $C_{gd1}$ :

- $C_{gd1}(1 - \frac{1}{A_0})$
- $C_{gd1}(1 - A_0)$

Node (1):

- $C_1 = C_{db7} + C_{gd7} + C_{sb5} + C_{gs5}$

Node (2):

- $C_2 = C_{gd5} + C_{db5} + C_{db3} + C_{gd3}$

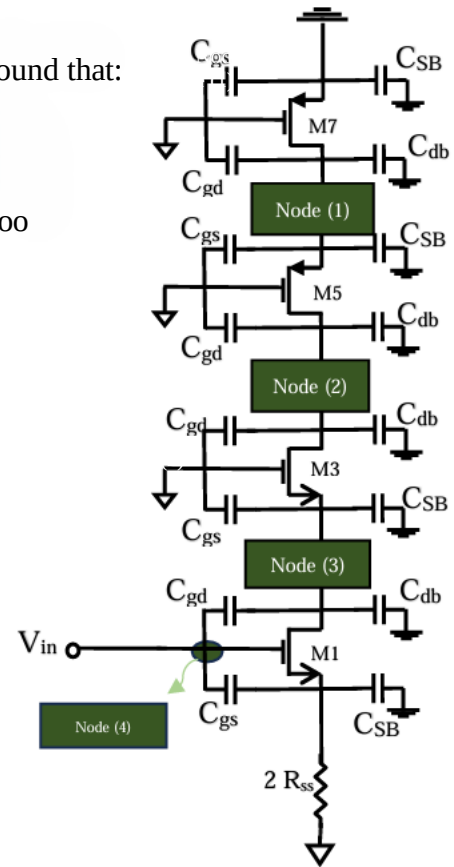
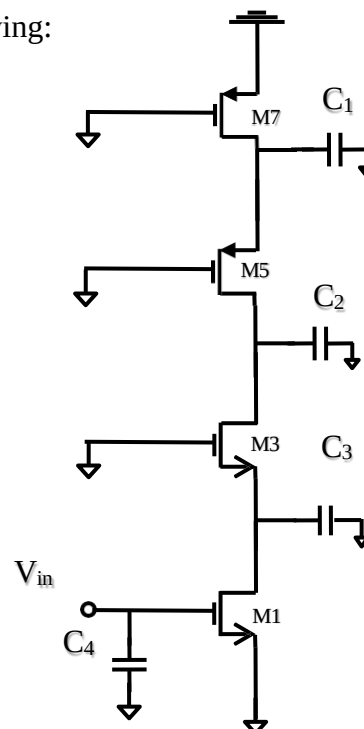
Node (3):

- $C_3 = C_{sb3} + C_{gs3} + C_{db1} + C_{gd1}(1 - \frac{1}{A_0})$

Node (4):

- $C_4 = C_{gs1} + C_{gd1}(1 - A_0)$

So, the Circuit reduced to the following:



$$A_0 = G_M|_{CS} \times R_{out} = \frac{-g_{m_1}}{1 + g_{m_1} R_s} \times R_{out} = -g_{m_1} R_{out} \quad ; R_s = 0$$

$$R_{out} = R_{LFD_1} // R_{LFS_3} = r_{0_1} (1 + g_{m_1} R_s) // \frac{1}{g_{m_3}} (1 + \frac{R_D}{r_{0_3}})$$

where,

$$R_D = \frac{r_{0_5}(1 + g_{m_5} R_{s_5})}{r_{0_5} g_{m_5} r_{0_7}} \quad ; R_{s_5} = r_{0_7}$$

$$R_{out} = r_{0_1} // \frac{1}{g_{m_3}} (1 + \frac{r_{0_5} g_{m_5} r_{0_7}}{r_{0_3}})$$

$$A_{v_0} = -g_{m_1} [ r_{0_1} // \frac{1}{g_{m_3}} (1 + \frac{r_{0_5} g_{m_5} r_{0_7}}{r_{0_3}}) ]$$

Now, we will get the poles and Zeros:

➤ Poles

$$\begin{aligned} \omega_{p_1} &= \frac{1}{C_1 R_1} \quad ; R_1 = R_{LFD_7} // R_{LFS_5} \\ \omega_{p_2} &= \frac{1}{C_2 R_2} \quad ; R_2 = R_{LFD_5} // R_{LFD_3} \\ \omega_{p_3} &= \frac{1}{C_3 R_3} \quad ; R_3 = R_{LFD_1} // R_{LFS_3} \\ \omega_{p_4} &= \frac{1}{C_4 R_4} = \infty \quad ; R_4 = 0 \end{aligned}$$

➤ Zeros

$$\omega_{z_1} = \left| \frac{g_{m_1}}{C_{gd_1}} \right|$$

Where,

$$R_{LFD_7} = r_{0_7} \rightarrow (1)$$

$$R_{LFS_5} = \frac{1}{g_{m_5}} \left( 1 + \frac{R_D}{r_{0_5}} \right) ; R_D = r_{0_3}(1 + g_{m_3}R_{S_3}) ; R_{S_3} = r_{0_1}$$

$$R_{LFS_5} = \frac{1}{g_{m_5}} \left( 1 + \frac{r_{0_3}g_{m_3}r_{0_1}}{r_{0_5}} \right) \approx \frac{r_{0_3}g_{m_3}r_{0_1}}{g_{m_5}r_{0_5}} \rightarrow (2)$$

$$R_{LFD_5} = r_{0_5}(1 + g_{m_5}R_{S_5}) ; R_{S_5} = r_{0_7}$$

$$R_{LFD_5} = r_{0_5}(1 + g_{m_5}r_{0_7}) \rightarrow (3)$$

$$R_{LFD_3} = r_{0_3}(1 + g_{m_3}R_{S_3}) ; R_{S_3} = r_{0_1}$$

$$R_{LFD_3} = r_{0_3}(1 + g_{m_3}r_{0_1}) \rightarrow (4)$$

$$R_{LFS_3} = \frac{1}{g_{m_3}} \left( 1 + \frac{R_D}{r_{0_3}} \right) ; R_D = r_{0_5}(1 + g_{m_5}R_{S_5}) ; R_{S_5} = r_{0_7}$$

$$R_{LFS_3} = \frac{1}{g_{m_3}} \left( 1 + \frac{r_{0_5}g_{m_5}r_{0_7}}{r_{0_3}} \right) \approx \frac{r_{0_5}g_{m_5}r_{0_7}}{g_{m_3}r_{0_3}} \rightarrow (5)$$

$$R_{LFD_1} = r_{0_1} \rightarrow (6)$$

Now, we will use the values from the simulation section

| Transistor | NMOS Input pair<br>(M1 & M2) | NMOS Active Loads<br>(M3 & M4) | Tail current source<br>(M9) |
|------------|------------------------------|--------------------------------|-----------------------------|
| $g_m$      | 693.0 $\mu s$                | 583.4 $\mu s$                  | 587.2 $\mu s$               |
| $r_o$      | 37.6 $K\Omega$               | 58.75 $K\Omega$                | 39.5 $K\Omega$              |
| $C_{gs}$   | 6.738 $fF$                   | 22.48 $fF$                     | 80.53 $fF$                  |
| $C_{gd}$   | 1.111 $fF$                   | 1.6 $fF$                       | 2.93 $fF$                   |
| $C_{sb}$   | 0.671 $fF$                   | 2.1 $fF$                       | 9.017 $fF$                  |
| $C_{db}$   | 0.011 $fF$                   | 0.0571 $fF$                    | 0.285 $fF$                  |

| Transistor | PMOS Active Loads<br>(M5 & M6) | PMOS Current Sources<br>(M7 & M8) |
|------------|--------------------------------|-----------------------------------|
| $g_m$      | 586.2 $\mu s$                  | 578.4 $\mu s$                     |
| $r_o$      | 69.6 $K\Omega$                 | 57.27 $K\Omega$                   |
| $C_{gs}$   | 107.6 $fF$                     | 0.137 $fF$                        |
| $C_{gd}$   | 5.333 $fF$                     | 6.577 $fF$                        |
| $C_{sb}$   | 12.05 $fF$                     | 15.4 $fF$                         |
| $C_{db}$   | 0.174 $fF$                     | 0.429 $fF$                        |

$$R_{LFD_7} = 57.27 \text{ K}\Omega$$

$$R_{LFD_3} = 1.35 \text{ M}\Omega$$

$$R_{LFS_5} = 3.15 \text{ K}\Omega$$

$$R_{LFS_3} = 68.2 \text{ K}\Omega$$

$$R_{LFD_5} = 2.4 \text{ M}\Omega$$

$$R_{LFD_1} = 37.6 \text{ K}\Omega$$

$$A_0 = -16.94$$

$$C_1 = 126.7 \text{ fF}$$

$$C_2 = 7.16 \text{ fF}$$

$$C_3 = 31.77 \text{ fF}$$

$$C_4 = 26.67 \text{ fF}$$

Now, we will get the poles and Zeros:

➤ Poles

$$\omega_{p_1} = \frac{1}{2\pi \times C_1 R_1} = 421.5 \text{ MHz}; R_1 = R_{LFD_7} // R_{LFS_5} = 2.98 \text{ k}\Omega$$

$$\omega_{p_2} = \frac{1}{2\pi \times C_2 R_2} = 25.73 \text{ MHz}; R_2 = R_{LFD_5} // R_{LFD_3} = 864 \text{ k}\Omega$$

$$\omega_{p_3} = \frac{1}{2\pi \times C_3 R_3} = 206 \text{ MHz}; R_3 = R_{LFD_1} // R_{LFS_3} = 24.23 \text{ k}\Omega$$

$$\omega_{p_4} = \frac{1}{C_4 R_4} = \infty; R_4 = 0$$

➤ Zeros

$$\omega_{z_1} = \left| \frac{g_{m_1}}{2\pi \times C_{gd_1}} \right| = 99.4 \text{ GHz}$$

# 5-DC Cascode Simulation

## 5.1 Schematic:

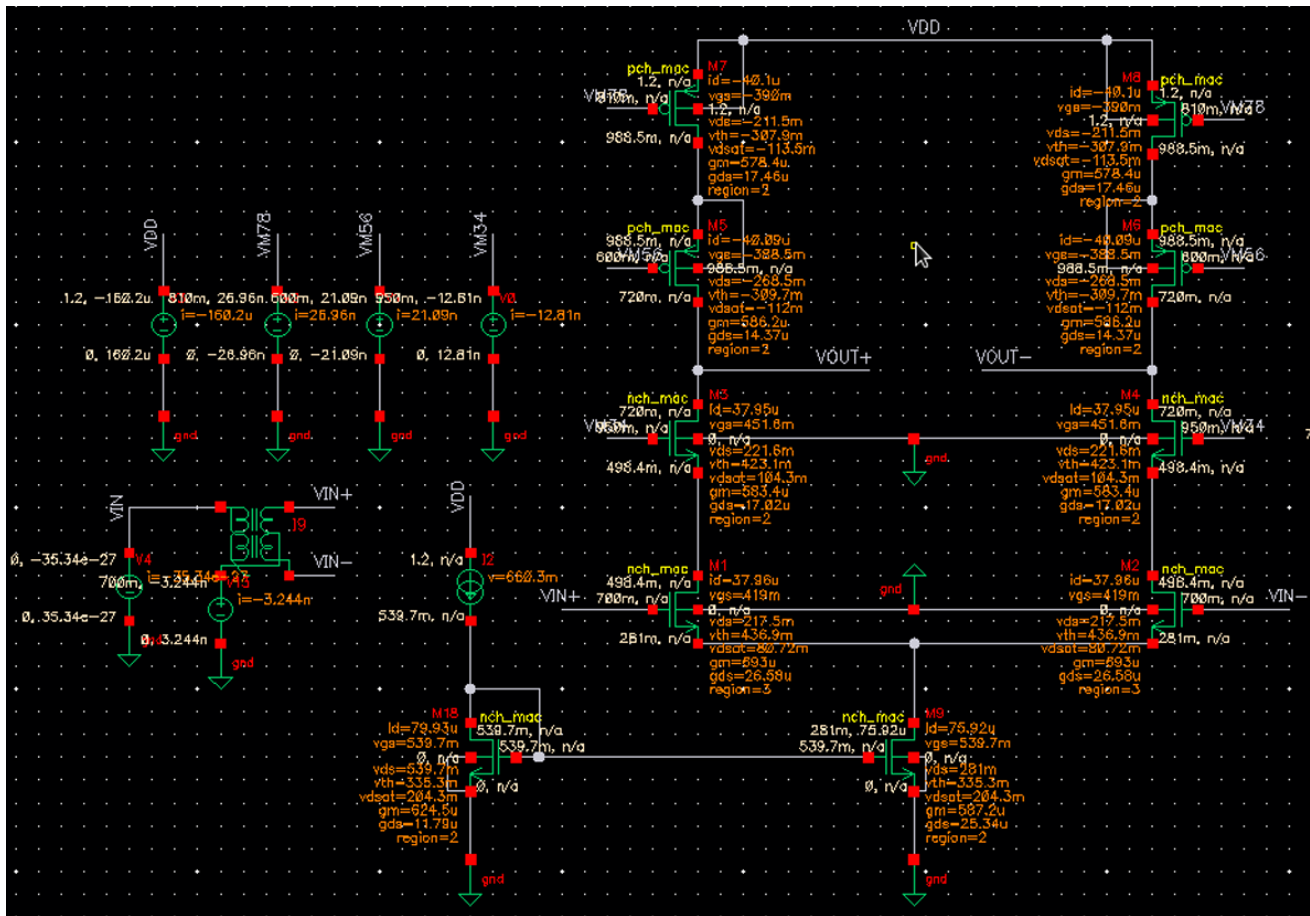


Figure 1 showing the schematic of the Telescopic cascode with current mirror.

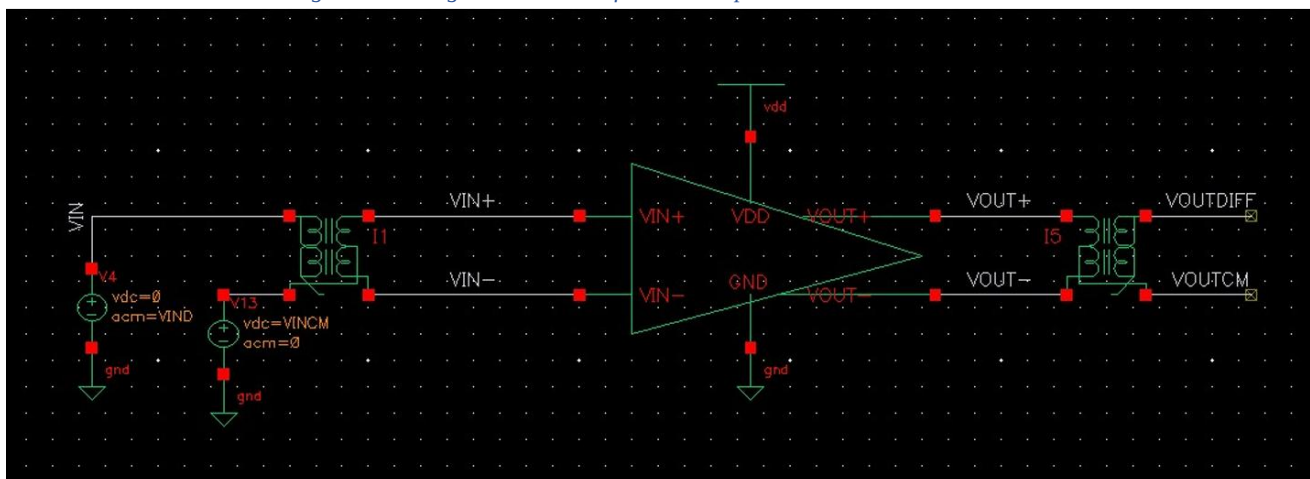


Figure 2 showing the Test bench implemented for the Telescopic cascode .

## 5.2 Setting Output Voltage value:

In a cascode analog circuit, the output is a **high impedance node**, making it an **ill-defined node** in terms of voltage. Since no low-impedance path sets its voltage, it must be externally defined to ensure proper circuit behavior. To address this issue, there are two solutions presented.

- **Common-Mode Feedback (CMFB)** circuits to regulate the output common-mode voltage.
- **Simulation technique** that defines the DC voltage without affecting AC performance.

In this case, a **simulation technique** was used. A resistive was connected between the output node and a common-mode voltage source (**VOCM**), with behavior dependent on the type of analysis as shown in figure 26.

- **In DC analysis**, the resistor acts as a **short circuit** (with resistance of 1 milli ohm), allowing the DC voltage at the output node to settle at VOCM.
- **In AC analysis**, the resistor is treated as an **open circuit** (with resistance of 1T ohm), ensuring it does not affect small-signal behavior.

By employing the simulation technique, the ill-defined Output node is correctly set using the VOCM source. Meanwhile, The AC simulation views the output as a high-impedance node, preserving the circuit's true small-signal characteristics. This approach is a practical simulation aid in Cadence environments. In real life case, CMFB circuitry is mandatory.

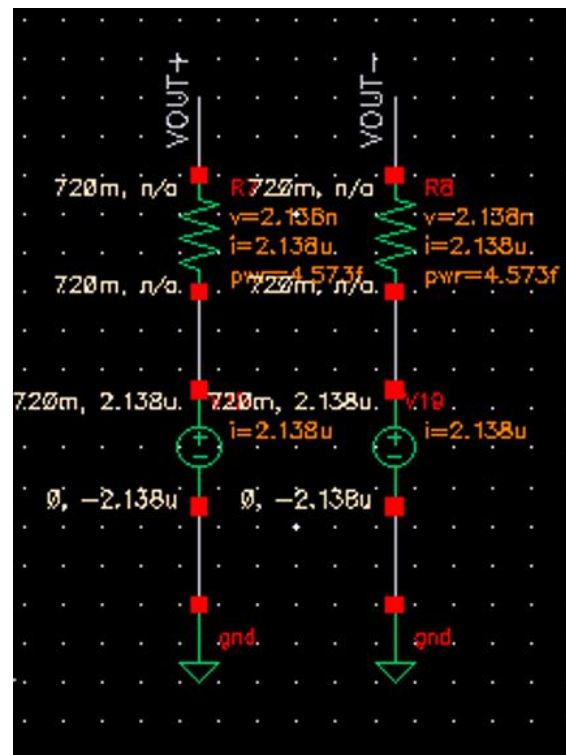


Figure 3 showing the resistance used to set voltage at output node

## 5.3 DC Operating point (OP):



Figure 4 showing the DC OP of Telescopic cascode.

From given values power, the total power of the circuit will be calculated.

$$\begin{aligned} P_{total} = \sum P_i &= (8.6 + 11.05 + 8.4 + 8.25) \times 2 + 21.33 \\ &= 93.93 \mu\text{W} \end{aligned}$$

Compare simulation results with hand calculations in a table.

| Transistor              | NMOS Input pair<br>(M1 & M2) | NMOS Active Loads<br>(M3 & M4) | Tail current source<br>(M9) |
|-------------------------|------------------------------|--------------------------------|-----------------------------|
| Simulated $g_m$         | 693 $\mu\text{s}$            | 583.4 $\mu\text{s}$            | 587.2 $\mu\text{s}$         |
| Simulated $I_d$         | 37.95 $\mu\text{A}$          | 37.95 $\mu\text{A}$            | 75.92 $\mu\text{A}$         |
| Simulated $V^*$         | 0.8 V                        | 0.104 V                        | 0.204 V                     |
| Simulated $r_{out}$     | 37.6 $K\Omega$               | 58.75 $K\Omega$                | 39.5 $K\Omega$              |
| Hand analysis $g_m$     | 720 $\mu\text{s}$            | 600 $\mu\text{s}$              | 400 $\mu\text{s}$           |
| Hand analysis $I_d$     | 40 $\mu\text{A}$             | 40 $\mu\text{A}$               | 80 $\mu\text{A}$            |
| Hand analysis $V_{ov}$  | 0.111 V                      | 0.1333 V                       | 0.2 V                       |
| Hand analysis $r_{out}$ | 45.2 $k\Omega$               | 69.15 $k\Omega$                | 100.72 $k\Omega$            |

| Transistor              | PMOS Active Loads<br>(M5 & M6) | PMOS Current Sources<br>(M7 & M8) |
|-------------------------|--------------------------------|-----------------------------------|
| Simulated $g_m$         | 586.2 $\mu\text{s}$            | 578.4 $\mu\text{s}$               |
| Simulated $I_d$         | 40.09 $\mu\text{A}$            | 40.09 $\mu\text{A}$               |
| Simulated $V^*$         | 0.112 V                        | 0.113 V                           |
| Simulated $r_{out}$     | 69.6 $K\Omega$                 | 57.27 $K\Omega$                   |
| Hand analysis $g_m$     | 600 $\mu\text{s}$              | 600 $\mu\text{s}$                 |
| Hand analysis $I_d$     | 40 $\mu\text{A}$               | 40 $\mu\text{A}$                  |
| Hand analysis $V_{ov}$  | 0.133 V                        | 0.133 V                           |
| Hand analysis $r_{out}$ | 70.27 $k\Omega$                | 75.79 $K\Omega$                   |

By comparing the simulation results with the hand analysis, only minimal deviations were observed across all transistor parameters, primarily caused by a slight current deviation from 40  $\mu\text{A}$  to 37.95  $\mu\text{A}$ .

#### Current Mismatch Explanation:

From the results, the PMOS drain current ( $I_{d,\text{PMOS}}=40.09 \mu\text{A}$ ) is slightly higher than the NMOS drain current ( $I_{d,\text{NMOS}}=37.95 \mu\text{A}$ ). This mismatch is attributed to the current flowing through the resistor used in the simulation testbench to fix the DC output common-mode voltage (VOCM). Since a closed-loop common-mode feedback (CMFB) circuit is not implemented, the output voltage is externally forced by the voltage source, which absorbs the difference between the sourcing current from the PMOS devices and the sinking current of the NMOS devices. This mechanism prevents the output node from drifting or railing despite the absence of CMFB.

However, a notable discrepancy appears between  $V_{ov}$  and  $V^*$  (which captures channel's inversion more accurately). This difference arises from the inapplicability of the square-law model in modern CMOS technologies, where short-channel effects and other non-idealities render the classical assumptions less accurate.

## 6- AC Cascode Simulation

Diff small signal:

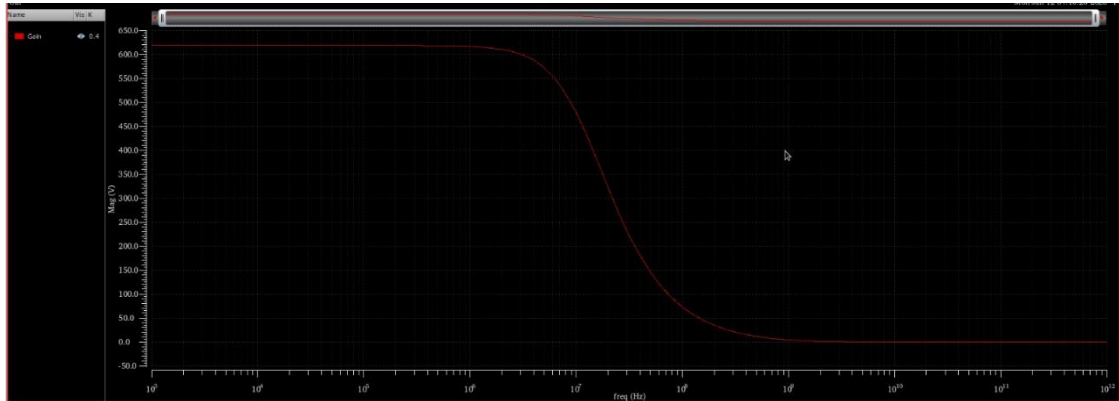


Figure 5 showing gain vs frequency for Telescopic cascode amplifier.

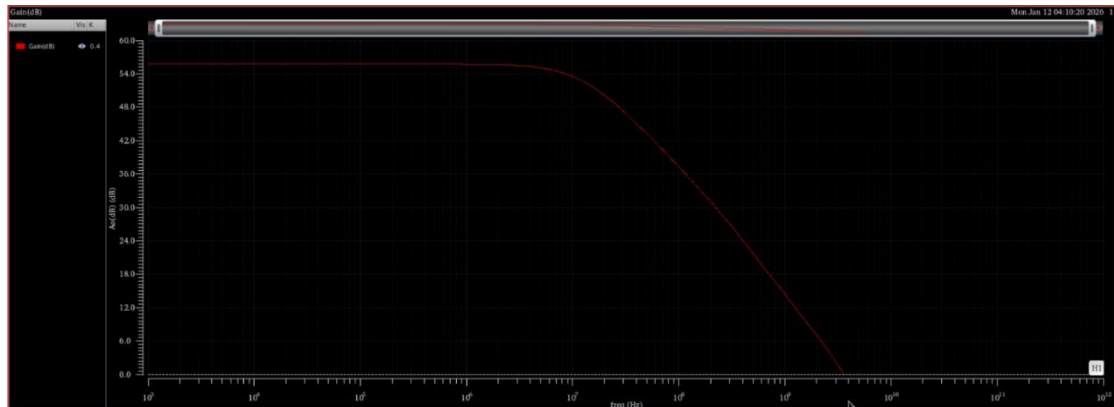


Figure 6 showing  $A_{vd_{dB20}}$  vs frequency for Telescopic cascode amplifier.

| Test                                       | Output   | Nominal | Spec  | Weight | Pass/Fail |
|--------------------------------------------|----------|---------|-------|--------|-----------|
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Gain     | ✓       |       |        |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | phase    | ✓       |       |        |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Gain(dB) | ✓       |       |        |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Ao       | 619.7   | > 500 |        | pass      |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Ao(dB)   | 55.84   | > 54  |        | pass      |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | BW       | 78.14M  |       |        |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | GBW      | 48.43G  |       |        |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Power    | 91.11u  | < 3m  |        | pass      |

Figure 7 showing the extracted simulation results of BW and  $A_{vd}$  from ADEXL.

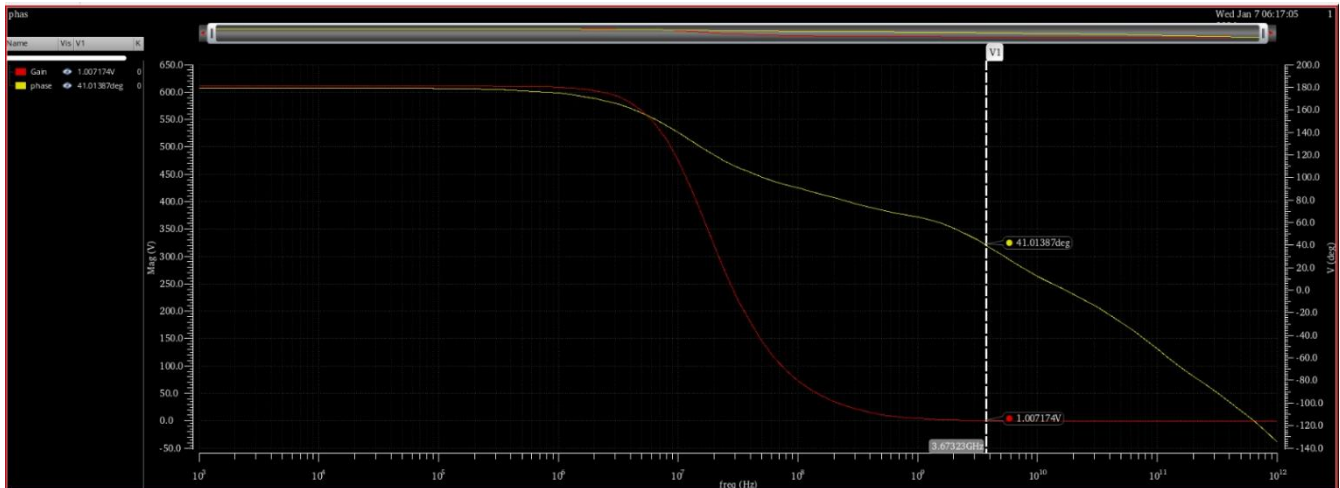
Compare simulation results with hand calculations in a table.

$$\begin{aligned}
 BW &= \frac{1}{C_{out} \times R_{out}} \\
 &= \frac{1}{(C_{gd5} + C_{db5} + C_{db3} + C_{gd3}) \times (r_{o5} \times g_{m5} \times r_{o7} \parallel r_{o3} \times g_{m3} \times r_{o1})} \\
 &= 1 / ((7.2 \times 10^{-15}) \times (2.34 \parallel 1.3) \times 10^3) =
 \end{aligned}$$

166.5 MHz

| Type       | $A_{vd}$ | BW        |
|------------|----------|-----------|
| Simulation | 619.7    | 78.14 MHz |
| Analytical | 620      | 166.5 MHz |

## 7- Phase Margin:



The amplifier exhibits a phase margin of approximately  $41^\circ$ , indicating stable operation with moderate ringing due to multiple high-frequency poles.

## 8- Corner and Monte Carlo Simulation

### 8.1 Corners Simulation:

To evaluate the robustness of the cascode amplifier design, a corners analysis was performed. The simulation included the fast-slow (fs), slow-fast(sf), slow (ss), and fast (ff) process corners. The primary design target was achieving a gain ( $A_v$ ) > 500.

By performing the corners simulation, the gain failed to meet the specification at the fast (ff) corner as shown in figure 8. The FF corner fails because the reduced output resistance of fast transistors significantly lowers the cascoded gain, causing the DC gain to fall below the required 500.

| Parameter    | Nominal        |  |  |  |  |  | Fast-Fast | Slow-Slow | Fast-Slow | Slow Fast |
|--------------|----------------|--|--|--|--|--|-----------|-----------|-----------|-----------|
| SaveOP.scs   |                |  |  |  |  |  |           |           |           |           |
| VDD          | 1.2            |  |  |  |  |  | 1.32      | 1.08      | 1.2       | 1.2       |
| temperature  | 27             |  |  |  |  |  | -40       | 125       | 27        | 27        |
| cm65gplus... | stat_mis       |  |  |  |  |  | ff        | ss        | fs        | sf        |
| cm65gplus... | tt_lvt         |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_rfm0s_25    |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_rfiivar     |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_dio_na25... |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_25ud18      |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_rfm0s       |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_25          |  |  |  |  |  | nom       | nom       | nom       | nom       |
| cm65gplus... | tt_dio_25ud18  |  |  |  |  |  | nom       | nom       | nom       | nom       |

| Test                                       | Output   | Nominal | Spec  | Weight | Pass/Fail | Min    | Max    | Fast-Fast | Slow-Slow | Fast-Slow | Slow Fast |
|--------------------------------------------|----------|---------|-------|--------|-----------|--------|--------|-----------|-----------|-----------|-----------|
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Gain     |         |       |        |           |        |        |           |           |           |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | phase    |         |       |        |           |        |        |           |           |           |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Gain(dB) |         |       |        |           |        |        |           |           |           |           |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Ao       | 619.7   | > 500 |        | fail      | 246.8  | 654.3  | 246.8     | 560.9     | 654.3     | 515.3     |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Ao(dB)   | 55.84   |       |        |           | 47.85  | 56.32  | 47.85     | 54.98     | 56.32     | 54.24     |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | BW       | 78.14M  |       |        |           | 59.07M | 155.5M | 155.5M    | 59.07M    | 78.29M    | 83.97M    |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | GBW      | 48.43G  |       |        |           | 33.13G | 51.23G | 38.38G    | 33.13G    | 51.23G    | 43.26G    |
| My_Projects:Telescopic_Cascode_Diff.Amp.:1 | Power    | 91.11u  | < 3m  |        | pass      | 89.5u  | 92.13u | 91.88u    | 89.5u     | 92.13u    | 89.94u    |

Figure 8 showing the extracted results of BW and  $A_{vd}$  for all corners (ff & ss & fs & sf) from ADEXL.

### 8.2 Monte Carlo Simulation:

To evaluate the statistical robustness and yield of the redesigned telescopic cascode amplifier, a Monte Carlo simulation was conducted using 1000 samples. The focus was on four key parameters: Gain-Bandwidth (GBW), Bandwidth (BW), voltage gain ( $A_v$ ) and power consumption.

Power Consumption Analysis:

As seen in figure 9, the power consumption results across all 1500 runs followed a gaussian distribution, with minimal spread. The statistical data is summarized below:

- **Mean power consumption: 91.1  $\mu$ W**
- **Standard deviation: 779.9 nW**

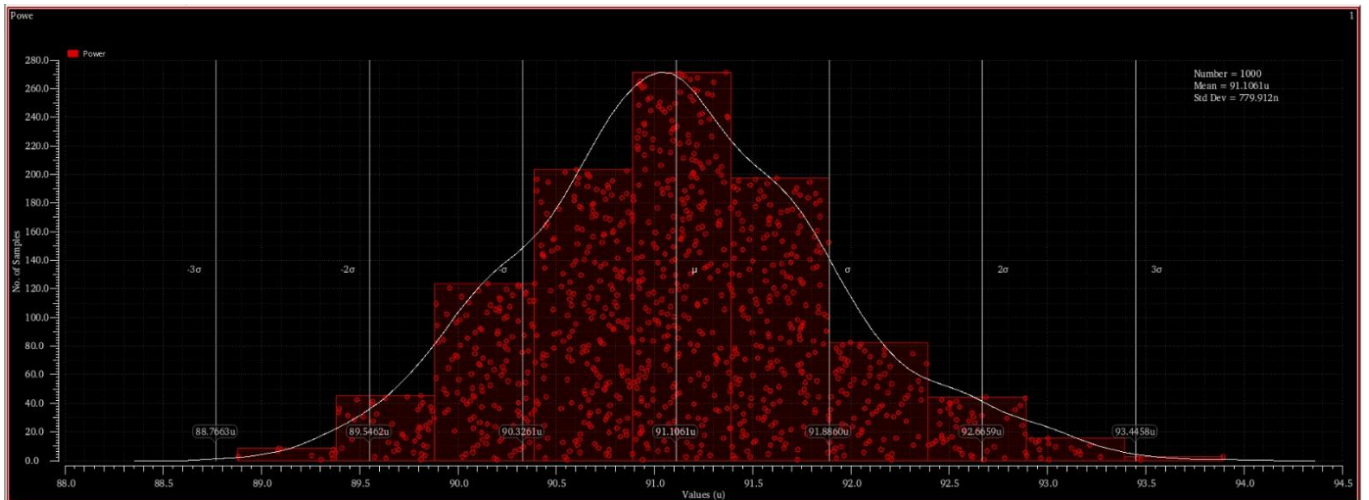


Figure 9 showing the statistical distribution of the total Power consumption under Monte Carlo simulation

All simulation runs met the power requirement, and the maximum observed value was 91.1  $\mu$ .

Gain Analysis:

The Monte Carlo histogram for gain, shown in figure 10, displays a Gaussian distribution:

- **Mean gain: 619.5**
- **Standard deviation: 2.89**

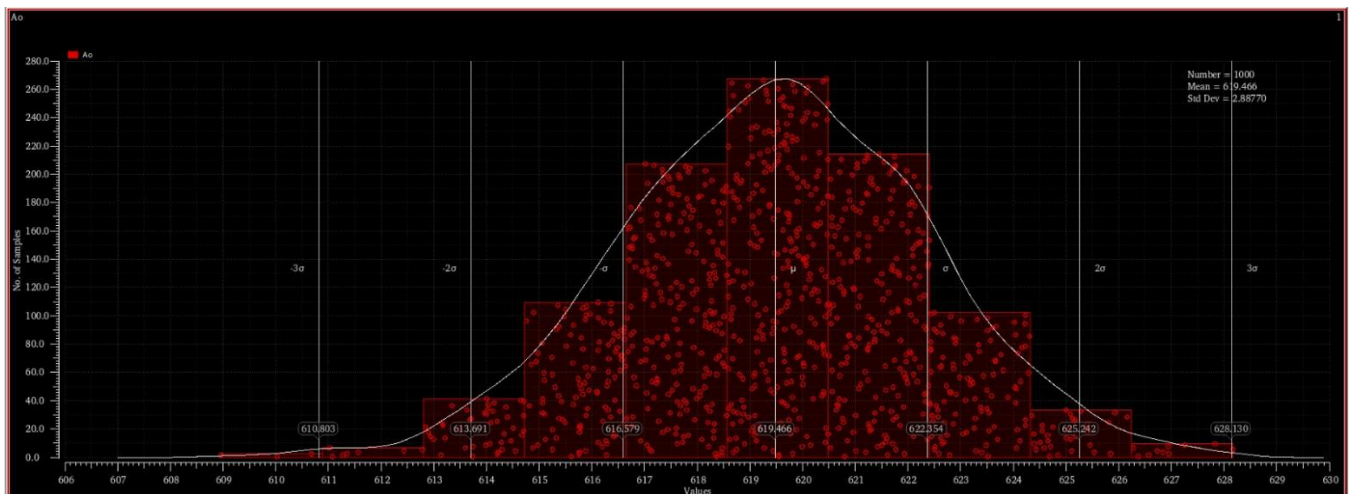
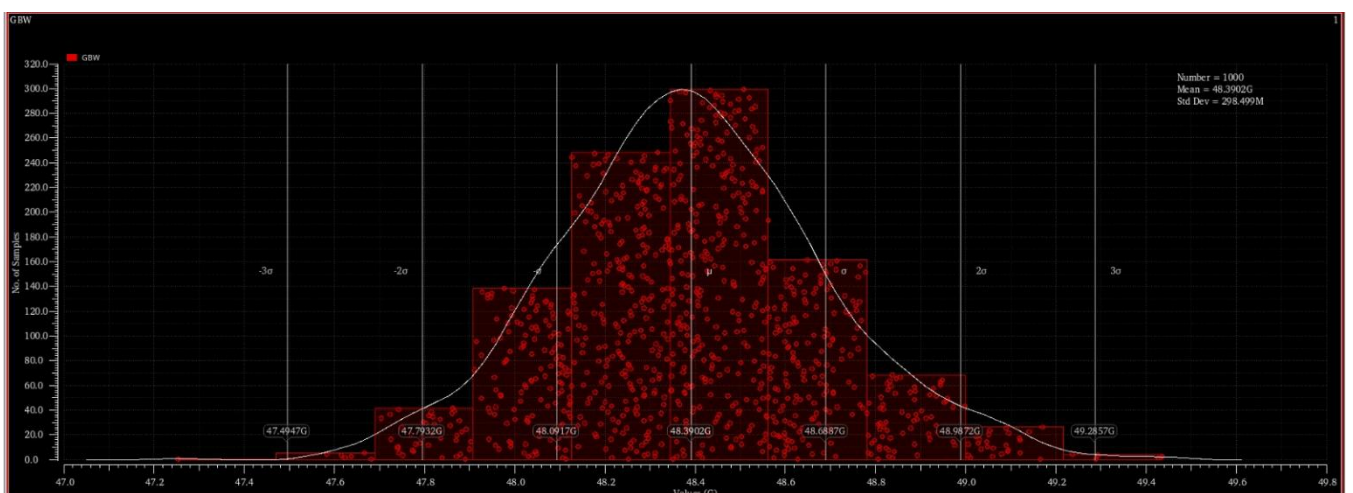
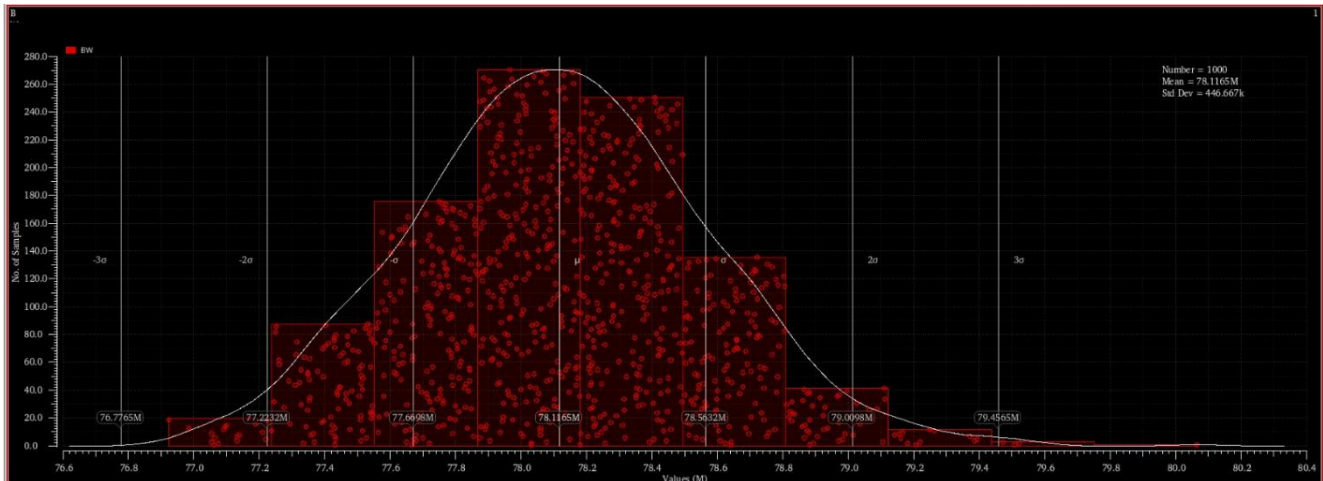


Figure 10 showing the statistical distribution of the  $A_v$  under Monte Carlo simulation

GBW Analysis:



## BW Analysis:



All simulation runs met the Gain requirement, and the minimum observed value was 609.

| Test                                                                                      | Name            | Yield | Min    | Target | Max    | Mean   | Std Dev | Cpk | Errors |
|-------------------------------------------------------------------------------------------|-----------------|-------|--------|--------|--------|--------|---------|-----|--------|
| Yield Estimate: 100 %(1000 passed/1000 pts) Confidence Level: <not set> Filter: <not set> |                 |       |        |        |        |        |         |     |        |
| - My_Projects:Telescopic_Cascode_Diff.Amp.:1                                              |                 |       |        |        |        |        |         |     |        |
| -                                                                                         | Ao(summary)     | 100   | 609    |        | 628.2  | 619.5  | 2.888   |     | 0      |
|                                                                                           | Ao              | 100   | 609    | info   | 628.2  | 619.5  | 2.888   |     | 0      |
| -                                                                                         | Ao(dB)(summary) | 100   | 55.69  |        | 55.96  | 55.84  | 40.5m   |     | 0      |
|                                                                                           | Ao(dB)          | 100   | 55.69  | info   | 55.96  | 55.84  | 40.5m   |     | 0      |
| -                                                                                         | BW(summary)     | 100   | 76.92M |        | 80.07M | 78.12M | 446.7k  |     | 0      |
|                                                                                           | BW              | 100   | 76.92M | info   | 80.07M | 78.12M | 446.7k  |     | 0      |
| -                                                                                         | GBW(summary)    | 100   | 47.25G |        | 49.44G | 48.39G | 298.5M  |     | 0      |
|                                                                                           | GBW             | 100   | 47.25G | info   | 49.44G | 48.39G | 298.5M  |     | 0      |
| -                                                                                         | Power(summary)  | 100   | 88.88u |        | 93.89u | 91.11u | 779.9n  |     | 0      |
|                                                                                           | Power           | 100   | 88.88u | info   | 93.89u | 91.11u | 779.9n  |     | 0      |

Figure 11 showing the summary of the Monte Carlo simulation over the two parameters:  $A_v$  and Power

## Conclusion:

The Monte Carlo analysis confirms that the designed circuit is highly reliable in terms of power consumption, but gain robustness still has room for improvement. With a 60% yield on gain, further enhancements such as sizing optimization or bias control could be explored to improve the design's tolerance to statistical variations.